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Advancing (Sparse) Multi-View Depth Estimation

Bachelor's Thesis

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Abstract

Zusammenfassung

Acknowledgements

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1

Introduction

Recovering the three-dimensional structure of a dynamic scene from visual observations is a foundational challenge in computer vision with direct implications for robotics and autonomous navigation [3]. Recent spatial foundation models—such as MAST3R-SfM [5], VGGT [8], PI3, and PI3X [10]—have achieved remarkable per-frame geometric fidelity by regressing dense 3D pointmaps directly from images. Yet in 4D settings, where temporal coherence across a video sequence is as critical as spatial accuracy, these models fail: assembled frame-by-frame, their outputs exhibit severe high-frequency jitter that renders them unsuitable for robotics applications requiring smooth, metric-consistent depth. Conversely, dedicated 4D architectures—VGGT4D [4] and MonoFusion [11]—sacrifice geometric precision to buy temporal smoothness, inflating Chamfer distance by 20–160 % and producing systematic alignment failures in sparse-view settings.

The central finding of this thesis is that a *decoupled* paradigm outperforms both families: take a strong per-frame 3D estimator, align its outputs into a common coordinate frame through a lightweight temporal alignment strategy, and apply a confidence-weighted, mask-aware Gaussian post-processing filter (Opt) to enforce static-scene consistency. This approach reduces jitter by 25–81 % across four models (MASt3R-SfM, VGGT, PI3, PI3X) while preserving geometry to within ± 2 –8 % Chamfer distance—a strictly superior trade-off to every native 4D architecture and geometry-refinement baseline we evaluate. At the two-camera extreme, VGGT combined with Opt beats the native 4D architecture VGGT4D on every single metric.

1.1 Background and Motivation

Classical multi-view reconstruction pipelines such as Structure-from-Motion (SfM) [7] produce accurate, metric-scale geometry in well-constrained settings: they rely on repeatable keypoint matching and bundle adjustment, which deliver precise camera poses and sparse point clouds when images share sufficient texture, overlap, and baseline. In sparse-view configurations of 2–4 cameras, however, these conditions frequently fail. Low inter-camera overlap means that few keypoints are visible in more than one view simultaneously [3]; short or degenerate baselines reduce parallax and destabilise triangulation [7]; and textureless surfaces—common on human skin, clothing, and manipulation objects—yield too few reliable correspondences for bundle adjustment to converge [9, 5]. The result is either outright reconstruction failure or incomplete, drift-prone geometry that cannot support downstream tasks such as grasping or object tracking.

Learning-based spatial models such as DUS3R [9], PI3, and PI3X [10] overcome many of these failure modes by directly regressing dense 3D pointmaps from image pairs, using learned priors to fill in low-texture and low-overlap regions where classical methods collapse. These models achieve strong per-frame accuracy—PI3 and PI3X in particular deliver the highest accuracy and completeness at a strict 1 cm robotics threshold among all models we evaluate. Their weakness is temporal: each frame is processed independently, with no mechanism to enforce consistency between consecutive reconstructions. The resulting frame-to-frame instability (jitter) can be large enough to make the output unusable for trajectory estimation or manipulation planning on a robotic platform.

The natural response is to incorporate temporal modelling. VGGT4D [4] extends spatial transformers with motion cues, and MonoFusion [11] fuses monocular depth priors across time. Both enforce sequence-level consistency, but at a cost we measure precisely: VGGT4D inflates Chamfer distance by 20–160 % relative to its per-frame baseline, and MonoFusion suffers systematic alignment convergence failures in sparse-view settings, producing duplicated subject artefacts rather than coherent reconstructions. Geometry post-processors such as GGPT [2] offer a third route, but function as unconstrained smoothers that destroy fine spatial structure, worsening Chamfer distance by 3–10×.

These results motivate the central design principle of this work: temporal consistency and geometric fidelity are not in fundamental conflict—they simply require a targeted rather than global smoothing strategy.

1.2 Problem Statement and Contributions

We address the problem of temporally consistent, geometrically precise 4D reconstruction from sparse multi-view video (2–4 cameras) of dynamic scenes, evaluated on the DEX-YCB [1] and Hi4D [12] datasets. The main contributions of this work are:

- **Unified 4D Benchmarking Framework.** A structured evaluation pipeline that subjects six base models—MASt3R-SfM, VGGT, PI3, PI3X, VGGT4D, and MonoFusion—to identical data-loading, semantic masking, and metric-computation conditions across DEX-YCB [1] and Hi4D [12]. The framework measures both 3D per-frame and 4D sequence-level performance via Chamfer distance, accuracy and completeness at a 1 cm threshold, temporal jitter, drift, and camera pose error, with static/dynamic decomposition where applicable (see Chapter 3). It additionally evaluates the effect of GGPT [2] geometry refinement applied on top of all base models except MonoFusion.
- **Comparative Temporal Alignment Study.** An analysis of three strategies for lifting per-frame 3D reconstructions into a globally consistent 4D representation (see Chapter 3). The Reference strategy aligns every frame to a common anchor using static-scene correspondences and the Umeyama similarity transform. The Hierarchical strategy builds a binary tree of pairwise alignments, propagating consistency bottom-up. The Pose-Graph Optimisation strategy computes all $\mathcal{O}(T^2)$ pairwise edges, weights them by alignment reliability, and refines all frame poses simultaneously via robust geodesic averaging on $\text{SO}(3)$. The Hierarchical strategy consistently delivers the best static-scene completeness and geometric stability; Pose-Graph Optimisation minimises the temporal consistency cost $\Delta_{\text{consistency}}$ (Equation 1.1) and best preserves dynamic-region accuracy.

$$\Delta_{\text{consistency}} = \text{Chamfer}_{4\text{D}} - \text{Chamfer}_{3\text{D}} \quad (1.1)$$

This quantity isolates the spatial degradation introduced purely by coordinate alignment, enabling direct comparison of alignment strategies and model families.

- **Confidence-Weighted Temporal Smoothing (Opt).** A ground-truth-free post-processing method that reduces frame-to-frame instability for any per-frame estimator without retraining (see ??). Opt applies a Gaussian temporal filter to already-aligned pointmaps, weighted

by model confidence and restricted to pixels classified as static by SAM2 [6]-derived masks—leaving dynamic pixels untouched to avoid suppressing real motion. A Pareto-frontier sweep over temporal bandwidth σ and blending strength α identifies a single robust configuration ($\sigma=4$, $\alpha=0.5$) that achieves a 25–81 % jitter reduction across MAST3R-SfM, VGGT, PI3, and PI3X while preserving geometry to within ± 2 –8 % Chamfer distance. At 2 views, VGGT with Opt surpasses VGGT4D on every metric, demonstrating that targeted post-processing of a strong spatial model outperforms native 4D architecture design under camera sparsity.

- **Empirical Characterisation of Architectural Families.** A systematic comparison of spatial-first models (PI3, PI3X, MAST3R-SfM, VGGT), native 4D architectures (VGGT4D, MonoFusion), and geometry-refinement post-processors (GGPT) reveals clear, quantifiable failure modes. PI3 and PI3X deliver the highest baseline spatial accuracy but suffer from the greatest temporal instability. MAST3R-SfM and VGGT offer competitive geometry with substantial high-frequency jitter that Opt largely eliminates. Native 4D models trade geometric precision for stability at a rate that never justifies the cost at any view count. GGPT is unsuitable for geometry-critical applications. Together these findings establish the decoupled framework—strong spatial estimator plus Opt—as the recommended approach for sparse-view robotic perception.

2

Literature Review and Background

3

Methodology: The Evaluation Framework

4

System Implementation

5

Results

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Refinement Strategy

7

Discussion

8

Conclusions

Bibliography

- [1] Yu-Wei Chao *et al.* “DexYCB: A Benchmark for Capturing Hand Grasping of Objects”. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2021.
- [2] Yutong Chen *et al.* “GGPT: Geometry-Grounded Point Transformer”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2026.
- [3] R. I. Hartley and A. Zisserman. *Multiple View Geometry in Computer Vision*. Second. Cambridge University Press, ISBN: 0521540518, 2004.
- [4] Yu Hu *et al.* *VGGT4D: Mining Motion Cues in Visual Geometry Transformers for 4D Scene Reconstruction*. 2025. arXiv: [2511.19971](https://arxiv.org/abs/2511.19971) [cs.CV]. URL: <https://arxiv.org/abs/2511.19971>.
- [5] Vincent Leroy, Yohann Cabon, and Jerome Revaud. *Grounding Image Matching in 3D with MAST3R*. 2024. arXiv: [2406.09756](https://arxiv.org/abs/2406.09756) [cs.CV].
- [6] Nikhila Ravi *et al.* “SAM 2: Segment Anything in Images and Videos”. In: *arXiv preprint arXiv:2408.00714* (2024). URL: <https://arxiv.org/abs/2408.00714>.
- [7] Johannes Lutz Schönberger and Jan-Michael Frahm. “Structure-from-Motion Revisited”. In: *Conference on Computer Vision and Pattern Recognition (CVPR)*. 2016.
- [8] Jianyuan Wang *et al.* *VGGT: Visual Geometry Grounded Transformer*. 2025. arXiv: [2503.11651](https://arxiv.org/abs/2503.11651) [cs.CV]. URL: <https://arxiv.org/abs/2503.11651>.
- [9] Shuzhe Wang *et al.* “DUSt3R: Geometric 3D Vision Made Easy”. In: *CVPR*. 2024.
- [10] Zihan Wang *et al.* “MonoFusion: Sparse-View 4D Reconstruction via Monocular Fusion”. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. Oct. 2025, pp. 8252–8263.

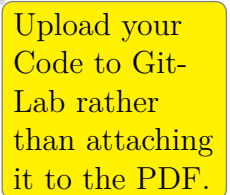
- [11] Zihan Wang *et al.* “MonoFusion: Sparse-View 4D Reconstruction via Monocular Fusion”. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. Oct. 2025, pp. 8252–8263.
- [12] Yifei Yin *et al.* “Hi4D: 4D Instance Segmentation of Close Human Interaction”. In: *Computer Vision and Pattern Recognition (CVPR)*. 2023.

List of Acronyms



Artifacts Availability

The artifacts associated with this thesis can be found in the ifi-GitLab¹. Within this repository, there are two main folders: one containing the code to fine-tune the model, and the other containing the code to generate the training data, including all the different datasets. The README file provided in the repository contains step-by-step instructions on how to set up the environment used in this thesis, install all necessary dependencies, and run the source code.



Upload your Code to Git-Lab rather than attaching it to the PDF.

¹<https://gitlab.ifi.uzh.ch/ddis/Students/Theses/<NAME>>



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Zürich, June 5, 2026

Signature of
Fabio Di Meo